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Human Dimensions of Pollinator Conservation: The Development and Testing of Survey Measures for Best Management Practice Adoption

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ABSTRACT

Pollinator conservation has been a topic of growing interest and concern, yet widespread adoption of pollinator best management practices (BMPs) on private lands has not occurred. This study was an empirical attempt to address the gap of landholder pollinator BMP adoption through two main objectives: 1) to develop and test a landholder survey constructed of multiple measurement scales based on the Theory of Planned Behavior (TPB), and 2) to use structural equation modeling to determine the interrelationships and influence of landholder TPB attributes on intentions to adopt pollinator BMPs. Analyses of two measurement models showed evidence of reliability and validity, and results from the structural models indicated landholders' perceived constraints had the strongest relative influence on their intentions to adopt pollinator BMPs. This study allowed for new understanding regarding landholders' involvement with pollinator conservation efforts, resulting in targeted engagement recommendations to support further participation in the future.

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Introduction

Pollinator conservation has become a topic of great interest and research around the world. Pollinator species provide the ecosystem service of pollination supporting approximately 35% of global food production and more than 80% of the world's flowering plant diversity (Klein et al. 2007; Ollerton, Winfree, and Tarrant 2011). Despite their beneficial impact, native insect pollinator species face population declines resulting primarily from habitat loss and increased chemical use (Kearns, Inouye, and Waser 1998; Kremen, Williams, and Thorp 2002; Potts et al. 2010; Ricketts et al. 2008). Urban expansion and increased agricultural intensification have resulted in greater homogenization of the landscape, contributing to the decline of mutualistic pollinator and plant systems (Garibaldi et al. 2017; Morandin, Long, and Kremen 2016; Williams et al. 2015). With growing concerns regarding food security and decreased biodiversity, people are attempting to find ways to conserve native pollinators. Considerable attention

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has been devoted to the development of best management practices (BMPs) and creation of funding opportunities in the United States (US) to support pollinator conservation efforts on both public and private lands (e.g., USDA FSA 2017; USDA FS 2015). Pollinator conservation efforts to provide suitable foraging and nesting habitat are crucial on private lands with over 60% of the lands in the US in private ownership (USDA 2018). Participatory and awareness opportunities have increased for landholders (i.e., landowner, manager, or renter), yet a gap remains between pollinator conservation promotion and BMP adoption (Garbach and Long 2017; Morandin, Long, and Kremen 2016).

Most prior pollinator research has focused on biological and ecological components, including pollinator importance in ecosystems, plant mutualisms and habitat requirements, and causes of population decline (e.g., Allen-Wardell et al. 1998; Kearns, Inouye, and Waser 1998; Losey and Vaughan 2006). Additional consequences of pollinator decline have also been investigated, including concerns of fitness and long-term persistence with reduced gene flow, negative impacts on obligate plant species, and food security concerns (e.g., Kovács-Hostyánszki et al. 2017; Potts et al. 2016; Sunderland 2011). Furthermore, recent studies have touched on the incorporation of pollinator habitat on private lands to provide myriad ecosystem benefits (e.g., Garibaldi et al. 2014; Morandin, Long, and Kremen 2016; Wratten et al. 2012). The culmination of these studies has allowed for the development of recommendations for landholder action, including the grouping of BMPs into three main categories: 1) establishment of forage habitat, 2) provision of nesting habitat, and 3) reduction of chemical use. These practices provide continuous forage for pollinator nectar and pollen needs and suitable nesting and larval host habitat for specialist species, while also promoting overall ecological health through the reduced use of chemicals (Mader et al. 2011). The implementation of these practices will depend on the land type, size, and management goal, requiring a process of individualization to fit landholder needs. Therefore, with the largest impact for pollinator conservation likely resulting from private landholder action, a conservation social science research focus is essential.

To explore dimensions of landholder conservation practice adoption, Diffusion of Innovation (DoI) theory and the Theory of Planned Behavior (TPB) have been used in numerous research studies (see Ajzen 1991; Fishbein and Ajzen 2015; Rogers 2003). Research focused on landholder knowledge and awareness has shown the potential impact on intention and adoption of certain conservation innovations (e.g., Bennett et al. 2017; Garbach and Long 2017; Greiner 2015). Attitudinal research has received considerable attention as several studies have shown a positive correlation between a favorable attitude toward a wildlife species or conservation practice and behavioral intentions toward a beneficial practice (e.g., Dayer et al. 2016; Greiner 2015; van Dijk et al. 2015). Nevertheless, inconsistent results from adoption research, including a lack of reliable factors that influence BMP adoption, underscore the complexity of social and environmental systems (Prokopy et al. 2019; Reimer et al. 2014). Additional attributes may prove to be more important regarding pollinator BMP adoption, such as constraints from using specific practices because of associated costs and required skills (Pannell et al. 2006). A study addressing the influence of multiple landholder attributes on adoption of pollinator BMPs is vital to further understanding and progress for pollinator conservation.

In this study, we sought to extend BMP adoption knowledge by addressing gaps in the human dimensions of pollinator conservation. Specifically, we aimed to focus on active pollinator BMPs that addressed the two primary reasons for decline of pollinator species. The purpose of this study was two-fold: 1) design and test a measurement instrument used to study attributes affecting landholder adoption of pollinator BMPs, and 2) use structural equation modeling to determine the interrelationships and influence of the four TPB attributes (i.e., *Attitudes*, *Injunctive Norms*, *Perceived Behavioral Control*, and *Intentions*). Based on previous research and a priori considerations, we developed five research hypotheses for testing: H1) Landholder attitudes differentiate between habitat and chemical pollinator BMPs, H2) *Attitudes* significantly and positively influences *Intentions*, H3) *Injunctive Norms* significantly and positively influences *Intentions*, H4) *Perceived Behavioral Control* significantly and positively influences *Intentions*, and H5) relative to the other TPB attributes, *Perceived Behavioral Control* has the strongest influence on *Intentions*.

Methods

Study Population

The agricultural and forestry production-dominated state of Mississippi was chosen for this study because of limited knowledge about landholders regarding pollinator conservation in the Southeast, compared to primary research regions such as the Midwest, and the variety of land use types. In Mississippi, roughly 90% of the land is privately-owned, with large sections in timber and agricultural production (MSU 2017; USDA 2018). Using a Mississippi tax record landowner database and the PROC SURVEYSELECT statement in SAS version 9.4, we selected a simple random sample of 4,000 Mississippi landholders with landholdings greater than or equal to 25 acres to receive the survey. The land size minimum of 25 acres was chosen primarily based on the substantial impact of larger sections of contiguous land for providing adequate pollinator foraging habitat and was believed to capture the majority of larger landholdings, while only reducing the overall sample population to 87.6% of the total properties listed in the tax database. Each landholder received a unique identifier to protect confidentiality of private records. All research protocols were approved by the Institutional Review Board for the Protection of Human Subjects prior to conducting the survey.

Questionnaire Development

To examine the differences of landholder attributes regarding adoption of pollinator BMPs, we developed a questionnaire to assess constructs related to the TPB and DoI theory, and to cover multiple landholder attributes and actual behaviors to better measure adoption per previous research recommendations (e.g., Miller 2017; Reimer et al. 2014). Although DoI constructs were included in the questionnaire, this paper's methods and results are focused solely on the TPB constructs as they were structurally modeled to determine influence on adoption intentions. To relate the TPB indicators to the nine pollinator BMPs, we developed nine indicators for each attribute construct (i.e., *Attitudes*, *Injunctive Norms*, *Perceived Behavioral Control*, and *Intentions*).

Researching attitudes, normative pressures, and constraints in this study would provide greater understanding of and potential opportunities for increasing pollinator BMP adoption rates. Studies have repeatedly shown attitudes to have a substantial effect on support for land management options on public and private lands (e.g., Fagan, Willcox, and Willcox 2018; Floress et al. 2017). In this study, we defined *Attitudes* as landholders' favorable or unfavorable evaluation of using pollinator BMPs on their property in the next year. Previous research has also indicated the importance of social interaction and identity, from the influence of a group of individuals to increase conservation action engagement through the use of social capital systems (i.e., norms, values, reciprocity), to an individual's desire to achieve pro-environmental status through impression management (i.e., revealing or concealing particular information to create a specific impression, such as being eco-friendly) and conspicuous conservation behaviors (e.g., Griskevicius, Tybur, and Van den Bergh 2010; Hargreaves 2016; Niemiec et al. 2018). To address social influences, we included an *Injunctive Norms* construct and defined it as landholders' normative pressures created by perceptions of others' beliefs that the landholders should use pollinator BMPs on their property in the next year. The "others" used for this construct included neighbors, those important to the landholders, and similar landholders. Prior work has shown constraints, such as an individual's perceptions of behavioral control related to their time, resources, and skills, to be influential on adoption behavior (e.g., Cary, Webb, and Barr 2001; Pannell et al. 2006). Thus, we used the *Perceived Behavioral Control* construct to determine potential adoption constraints faced by landholders. We defined this constraint construct as landholders' degree of belief that they have the skills, resources, and time to use pollinator BMPs on their property in the next year.

The dependent construct for this study was *Intentions*. Although intentions do not guarantee action, responses were necessary to determine attribute influence on the landholders' decision making. We defined *Intentions* as the landholders' plan regarding adoption of pollinator BMPs on their property in the next year. The simultaneous measurement of these constructs allowed for the determination of evidence of their reliability, validity, and structural relationships while maintaining the theoretical foundation purported by previous studies (e.g., Ajzen 1991; Fishbein and Ajzen 2015). Each construct had multiple indicators measured with 5-point Likert scales with response format 1 = strongly disagree, 2 = disagree, 3 = neutral, 4 = agree, and 5 = strongly agree.

The initial questionnaire was reviewed by expert judges (i.e., six university professors versed in survey research) for face and content validity and pretested with a random sample of 50 Mississippi landholders in April 2018. Minor modifications related to question wording were made to the questionnaire before the final instrument was disseminated in June 2018. In addition to the questionnaire, we developed and provided a leaflet with reference information for the landholder including examples of insect pollinator types, nectar and host plant species, and information about the nine pollinator BMPs selected for the study. The nine BMPs encompassed six habitat practices (i.e., planting nectar species, using cover crops, creating field borders, planting host plant species, creating brush piles, and using conservation tillage) and three chemical practices (i.e., using Integrated Pest Management, using targeted herbicides, and adjusting pesticide application timing).

Survey Implementation and Response

We used Dillman's (2000) Tailored Design Method for the dissemination of the finalized survey to Mississippi landholders. The survey was initiated in June 2018 to avoid the majority of planting and harvesting efforts for agricultural landholders in Mississippi. A cover letter, questionnaire, and postage-paid Business Reply envelope (i.e., complete packet) was initially sent (Day 1), followed by a postcard reminder one week later (Day 8). Second and third complete packets were sent as necessary on Day 21 and Day 42 to non-respondents (Dillman 2000). No incentives were used for this survey. Responses were accepted up to 90 days after the initiation of the survey. Data were entered into a Microsoft Access database and double-checked to avoid transcriptional error (Hunter and Schmidt 1990).

A total of 1,429 Mississippi landholders responded to the survey. After correcting for non-deliverables ($n = 286$), a 38.5% effective response rate was attained. We removed respondents who were non-eligibles ($n = 505$; e.g., those who were no longer managing or holding the property ($n = 109$), were reported as deceased ($n = 88$), refused the survey ($n = 231$), or were suspected of acquiescence or response set bias ($n = 77$)), resulting in a final usable total of 924 respondents. To determine acquiescence or response set bias, we used methods proposed by Collier (2020) to calculate the standard deviation of the Likert scale questions and removed respondents with scores less than 0.50 due to little or no variance of responses across the survey. We used non-response techniques to identify potential non-response bias (Fisher 1996) with known variables in the landholder database: 1) resident/non-resident, 2) rural/urban county, and 3) land size in acres. All tests were non-significant ($p > 0.05$), indicating response was not biased based on these variables.

The majority of respondents were older ($M = 66.38$ years, $SD = 11.00$, median = 67.00), white (89.0%), males (72.0%). They held an average property size of 145.61 acres ($SD = 266.92$, median = 70.80) and the plurality (40.0%) indicated timber production was their property's primary land use type, followed by animal agriculture (18.7%), residence (16.4%), recreation/leisure (14.7%), and crop agriculture (7.6%). Most (63.7%) respondents indicated they had adopted at least one pollinator BMP on their surveyed property, while few (3.9%) reported they were using pollinator BMPs on other landholdings.

Data Analysis

The data analysis for this study included an exploratory approach to test for evidence of reliability and validity of the measurement instrument, and a confirmatory approach to confirm the evidence and investigate the structural relationships of the TPB attributes. Prior to conducting analyses, we addressed missing values using a regression imputation in IBM SPSS Statistics version 26 (SPSS) based on prior research recommendations (Hair et al. 2009). Using SAS version 9.4, we randomly split the total sample into two subsamples of 462 respondents each to allow for primary development and cross-checking (DeVellis 2012), with the first used as an exploratory sample (i.e., Sample 1) in initial exploratory and confirmatory analyses

and the second used as the confirmatory sample (i.e., Sample 2) in additional measurement and structural analyses.

We conducted reliability and exploratory factor analyses (EFA) in SPSS to analyze Sample 1's data for evidence of reliability and validity. Reliability analyses included a review of Cronbach's alpha (α), inter-item and item-total correlations, and squared multiple correlations. For validity and dimensionality, we conducted a dimension reduction factor analysis using a principal axis factoring extraction based on eigenvalues greater than 1.00 and a Varimax rotation as a rigorous test of the items maximizing loading variance to produce a simple loading pattern and facilitate factor interpretability (DeVellis 2012; Netemeyer, Bearden, and Sharma 2003). We then used Sample 1 to conduct a confirmatory factor analysis (CFA) in IBM SPSS AMOS version 25 (AMOS) to test the measurement model before using Sample 2 in further analyses.

We used the measurement model to further test reliability and validity of the measures. To test reliability, we calculated a composite reliability for each construct (Hair, Ringle, and Sarstedt 2011). To test validity, we calculated construct average variance extracted (AVE) values to determine if they met the 0.50 threshold for convergent validity, and shared variance values to conclude if there was evidence of discriminant validity (Fornell and Larcker 1981; Hair, Ringle, and Sarstedt 2011). Per previous research recommendations, we reviewed multiple model fit statistics to determine initial fit to avoid incorrect interpretation based solely on reviewing the chi-square statistic as it is sample size sensitive (Bagozzi and Yi 2012; Chin, Peterson, and Brown 2008; Garver and Mentzer 1999). Cutoff values for model fit acceptability are difficult to designate and have varied, with many researchers referring to conventional rules of thumb (Hu and Bentler 1999). For this study, the models would have acceptable fit with cutoff values less than or equal to 0.08 for root mean square residual (RMR) and 0.10 for root mean square error of approximation (RMSEA), and greater than or equal to 0.90 for comparative fit index (CFI) and Tucker-Lewis index (TLI), with more stringent values closer to 0.06 (RMSEA) and 0.95 (CFI & TLI) preferred based on previous recommendations (Hu and Bentler 1999; Netemeyer, Bearden, and Sharma 2003). Measurement model modification indices were also reviewed to indicate potential model improvements. To avoid capitalizing on chance (Bagozzi and Yi 2012), no model edits were made after moving forward to test the measurement and structural models with Sample 2.

As the data were collected at one time, we assessed common method bias to determine if there was an inflation of correlations between dependent and independent variables (Podsakoff et al. 2003). We performed a measurement model analysis using an unmeasured latent construct with paths to each measurement indicator from *Attitudes*, *Injunctive Norms*, *Perceived Behavioral Control*, and *Intentions*. We examined chi-square values between this model and the original to determine if the difference was significant (i.e., $\chi^2/1df \geq 3.84$; Podsakoff et al. 2003). We also performed an invariance test to ensure response differences were not entirely the result of survey variance (Steenkamp and Baumgartner 1998). For the structural model, we performed model fit analyses to determine fit and reviewed structural relationships to test the hypothesized relationships.

Results

Sample 1's data showed initial evidence of reliability for each construct (i.e., each Cronbach's $\alpha > 0.89$). The EFA revealed unidimensionality for the *Perceived Behavioral Control* and *Intentions* constructs. It also showed two dimensions underlying the *Attitudes* construct, with one factor containing the habitat pollinator BMP indicators and one factor containing the chemical indicators, supporting Hypothesis 1. However, the Integrated Pest Management (IPM) indicator split between the two factors and loaded more highly on the habitat factor. Additional statistical review of this indicator resulted in it being dropped from the *Attitudes* construct in further analysis. The *Injunctive Norms* construct also showed respondent differentiation and two underlying factors; thus, the habitat and chemical indicators were further analyzed and modeled separately to isolate their effects in the structural relationships. One of the habitat indicators (i.e., creation of brush piles) was dropped from each construct as it failed to meet respective statistical thresholds.

Sample 1's CFA for the habitat model showed all indicators loading on their respective underlying constructs. A higher correlation value was observed between two habitat indicators related to planting (i.e., planting nectar species and planting host plant species; $r = 0.68$); thus, the latter was dropped from further analysis because of potential redundancy (Clark and Watson 1995). Modification indices indicated valid opportunities for model improvement, with which acceptable model fit was obtained. The chi-square statistic was significant ($\chi^2 = 388.216$, $df = 93$, $p < 0.001$); however, because this statistic is sample size sensitive, we reviewed additional fit indices (Bagozzi and Yi 2012). The model was determined to have acceptable fit with a review of RMR (0.04), RMSEA (0.08), CFI (0.94), and TLI (0.93). Sample 1's CFA for the chemical model also showed all indicators loading on their respective constructs and there were no valid modification indices suggestions. The chi-square statistic was also significant ($\chi^2 = 193.47$, $df = 38$, $p < 0.001$); however, acceptable model fit was determined based on RMR (0.04), RMSEA (0.09), CFI (0.96), and TLI (0.94). The habitat and chemical measurement models were analyzed again with Sample 2 before moving into the structural models.

Sample 2's CFAs for both the habitat and chemical models showed evidence of reliability and all indicators loaded on their respective underlying constructs (Tables 1 and 2). Each construct AVE exceeded 0.50, providing evidence of convergent validity (Table 3; Hair, Ringle, and Sarstedt 2011). Discriminant validity was supported as no shared variance between constructs exceeded the individual construct AVEs (Fornell and Larcker 1981; Hair, Ringle, and Sarstedt 2011). Both measurement models had at least acceptable fit based on several fit indices. For the habitat model, the chi-square statistic was significant ($\chi^2 = 477.55$, $df = 93$, $p < 0.001$), yet model fit was deemed acceptable with a review of multiple fit statistics (Table 1). For the chemical model, the chi-square statistic was also significant ($\chi^2 = 126.58$, $df = 38$, $p < 0.001$); however, good model fit was determined based on the additional fit statistics (Table 2). Common method bias was not a substantial concern for either model (*habitat*: $\chi^2/1df = 0$; *chemical*: $\chi^2/1df = 3.03$). Finally, the invariance test between males and females confirmed configural and at least partial metric invariance for both models (*habitat*: $\Delta df = 12$, $\chi^2 = 20.49$, $p = 0.058$; *chemical*: $\Delta df = 6$, $\chi^2 = 8.18$, $p = 0.225$).

Table 1. Confirmatory factor and reliability analyses for habitat indicators.

Constructs	Standardized Factor Loading	t value
<i>Attitudes</i> (C.R. = 0.80)		
Please indicate the extent to which you agree or disagree with each of the following statements regarding pollinator habitat on this property:		
(ATT1) I think planting nectar species on this property would improve the health of the property	0.73	**
(ATT2) I think creating field borders on this property would be beneficial for wildlife	0.75	13.42
(ATT3) I think using cover crops on this property would help reduce negative impacts on the environment	0.74	13.32
(ATT4) I think practicing conservation tillage on this property would help reduce negative impacts on the environment	0.63	10.33
<i>Perceived Behavioral Control</i> (C.R. = 0.86)		
Please indicate the extent to which you agree or disagree with each of the following statements, ending each statement with "... on this property in the next year":		
(PBC1) I have the skills needed to plant nectar species	0.77	**
(PBC2) I have the resources to use cover crops	0.83	17.45
(PBC3) I have the time to create field borders	0.80	16.92
(PBC4) I have the resources to practice conservation tillage	0.71	14.55
<i>Injunctive Norms</i> (C.R. = 0.90)		
Please indicate the extent to which you agree or disagree with each of the following statements, ending each statement with "... property in the next year":		
(IN1) Those important to me think I should establish forage habitat on this	0.96	**
(IN2) Landholders like me think I should establish forage habitat on this	0.84	20.96
(IN3) My neighbors think I should provide nesting habitat on this	0.71	14.02
(IN4) Those important to me think I should provide nesting habitat on this	0.82	15.95
<i>Intentions</i> (C.R. = 0.90)		
Please indicate the extent to which you agree or disagree with each of the following statements, ending each statement with "... in the next year":		
(INT1) I will plant nectar species on this property	0.78	**
(INT2) I will use cover crops on this property	0.90	21.64
(INT3) I will create field borders on this property	0.83	19.56
(INT4) I will practice conservation tillage on this property	0.82	19.20

Note: Model Fit Statistics: $\chi^2 = 477.55$; $df = 93$; CFI = 0.92, TLI = 0.90, RMR = 0.04, RMSEA = 0.09.

**Items constrained for identification purposes.

C.R.: composite reliability; CFI: comparative fit index; TLI: Tucker-Lewis index; RMR: root mean square residual; RMSEA: root mean square error of approximation.

The structural analysis results showed both models fit the data relatively well (Table 4). After establishing model fit, we reviewed the hypothesized relationships. For both models, the hypothesized relationships were found to be significant and supported (Figure 1). *Attitudes* was significantly and positively related to *Intentions* (Hypothesis 2), as was *Injunctive Norms* (Hypothesis 3) and *Perceived Behavior Control* (Hypothesis 4). Relative to the other constructs, *Perceived Behavioral Control* had the strongest relationship to *Intentions* (*habitat*: $\gamma = 0.55$; *chemical*: $\gamma = 0.52$; Hypothesis 5; Table 4). Although the *Attitudes* standardized estimates were the same between the habitat and chemical models, the *Injunctive Norms* standardized estimate was slightly higher for the chemical than for the habitat model (*habitat*: $\gamma = 0.20$; *chemical*: $\gamma = 0.22$) and the *Perceived Behavioral Control* standardized estimate was slightly lower as previously reported. We

Table 2. Confirmatory factor and reliability analyses for chemical indicators.

Constructs	Standardized Factor Loading	<i>t</i> value
<i>Attitudes</i> (C.R. = 0.87)		
Please indicate the extent to which you agree or disagree with each of the following statements regarding pollinator habitat on this property:		
(ATT5) I think using targeted herbicides on this property would help reduce negative impacts on the environment	0.77	**
(ATT6) I think adjusting pesticide application timing on this property would be beneficial for wildlife	0.98	11.39
<i>Perceived Behavioral Control</i> (C.R. = 0.90)		
Please indicate the extent to which you agree or disagree with each of the following statements, ending each statement with "... on this property in the next year":		
(PBC5) I have the skills needed to use IPM practices	0.80	**
(PBC6) I have the resources to use targeted herbicides	0.91	22.28
(PBC7) I have the time to adjust pesticide application timing	0.89	21.84
<i>Injunctive Norms</i> (C.R. = 0.93)		
Please indicate the extent to which you agree or disagree with each of the following statements, ending each statement with "... property in the next year":		
(IN5) My neighbors think I should reduce my chemical use on this	0.88	**
(IN6) Those important to me think I should reduce my chemical use on this	0.96	29.51
(IN7) Landholders like me think I should reduce my chemical use on this	0.86	25.20
<i>Intentions</i> (C.R. = 0.92)		
Please indicate the extent to which you agree or disagree with each of the following statements, ending each statement with "... in the next year":		
(INT5) I will use IPM practices on this property	0.82	**
(INT6) I will use targeted herbicides on this property	0.91	24.77
(INT7) I will adjust pesticide application timing on this property	0.96	26.18

Note: Model Fit Statistics: $\chi^2 = 126.58$; $df = 38$; CFI = 0.98, TLI = 0.97, RMR = 0.03, RMSEA = 0.07;

**Items constrained for identification purposes; IPM: Integrated Pest Management; C.R.: composite reliability; CFI: comparative fit index; TLI: Tucker-Lewis index; RMR: root mean square residual; RMSEA: root mean square error of approximation.

used McQuitty's (2004) criteria for statistical power calculations and found the power of the model to be > 0.99 , providing adequate support of the results. Additionally, we calculated effect sizes for the effect of previous adoption behavior on habitat *Intentions* ($t(460) = 6.564$, $p < .001$, Cohen's $d = .628$) and chemical *Intentions* ($t(460) = 5.730$, $p < .001$, Cohen's $d = 0.554$). These results indicate previous adoption behavior has at least a medium effect (Cohen 1988); thus, differences in BMP adoption intentions would be apparent when comparing those who have and have not previously adopted pollinator BMPs.

Discussion

Private landholders are an integral part of the conservation puzzle; therefore, to increase the impact of conservation efforts, one must go beyond consideration of the species or habitat of concern to incorporate social science aspects into research. Although there have been vital research efforts investigating landholder attitudes toward pollinator importance, willingness-to-pay for the pollination ecosystem service, and intentions

Table 3. Correlations, means, standard deviations, and average variance extracted for the habitat and chemical models.

Construct	Mean (SD)	AVE	Intercorrelation of constructs			
Habitat model			1	2	3	4
1. Attitudes	3.60 (0.69)	0.51	1.00			
2. Perceived Behavioral Control	2.79 (0.95)	0.61	0.35	1.00		
3. Injunctive Norms	2.84 (0.75)	0.70	0.33	0.35	1.00	
4. Intentions	2.83 (0.88)	0.70	0.49	0.65	0.47	1.00
Chemical model						
1. Attitudes	3.53 (0.77)	0.78	1.00			
2. Perceived Behavioral Control	2.68 (0.95)	0.76	0.29	1.00		
3. Injunctive Norms	2.60 (0.83)	0.81	0.13	0.13	1.00	
4. Intentions	2.83 (0.89)	0.80	0.41	0.59	0.32	1.00

Note: SD: standard deviation; AVE: average variance extracted.

toward practices that may provide pollinator habitat in the US and elsewhere (e.g., Breeze et al. 2015; Garbach and Long 2017; Munyuli 2011), a gap remains regarding how certain landholder attributes influence intentions to adopt specific pollinator BMPs. Our study was an empirical attempt to address this gap by developing and testing a measurement instrument that can be used to survey and understand landholder attitudes, normative pressures, and constraints from adopting pollinator BMPs.

Consistent with previous TPB conservation practice research (e.g., Borges and Lansink 2016; Roesch-McNally, Arbuckle, and Tyndall 2017), three of the TPB attributes were shown to have significant influence on *Intentions*, indicating favorable attitudes toward practices, presence of normative pressures, and positive perceptions of one's behavioral control all positively influence a landholder's intention to adopt pollinator BMPs. Although a large component of adoption research has been focused on individuals' attitudes as they have been believed to be the strongest predictor of behavioral intentions, our study showed *Perceived Behavioral Control* to have the strongest relative influence on *Intentions*. This indicates perceptions of constraints, such as time, resources, and skills, may be a stronger predictor of an individual's decision to adopt pollinator BMPs than either having a favorable or unfavorable attitude or feeling normative pressures from others to adopt. These findings suggest that to increase intentions to adopt pollinator BMPs overall, it will be important to consider more than just landholders' attitudes in future pollinator BMP adoption research.

Pollinator conservation has become a buzz phrase over the past decade as nonprofit organizations and federal agencies have increased efforts to spread awareness and encourage involvement; however, awareness is not synonymous with knowledge or skills. The results of our study indicate for habitat and chemical BMPs, landholders would be better supported in BMP adoption if they felt more empowered to take the actionable steps necessary to implement the practices. Increased empowerment could be attained from more access to workshops or demonstration areas to build landholder capacity and confidence through increased knowledge, skills, and recognition of the time required for the practices (Niles, Brown, and Dynes 2016; Pannell et al. 2006). Additionally, increased knowledge of and connections to funding opportunities to support these practices on their land would benefit landholders. Furthermore, understanding what practices are feasible and compatible for a landholder is also essential. For example, none of the top five agricultural enterprises for Mississippi (i.e., poultry and eggs, forestry, soybeans, cotton, and corn; MDAC 2019) require the ecosystem service

Table 4. Structural model results for the habitat and chemical models.

<i>Hypothesized relationships</i>	Habitat standardized estimates (<i>t</i> values)	Chemical standardized estimates (<i>t</i> values)	Hypothesis supported
H2: Attitudes significantly and positively influences Intentions	0.25 (5.39)	0.25 (6.25)	Supported
H3: Injunctive Norms significantly and positively influences Intentions	0.20 (4.78)	0.22 (5.77)	Supported
H4: Perceived Behavior Control significantly and positively influences Intentions	0.55 (10.48)	0.52 (11.16)	Supported
<i>Squared Multiple Correlation</i> Intentions	Habitat 0.66	Chemical 0.52	
<i>Model Fit Statistics</i>			
Habitat: $\chi^2 = 477.55$, $df = 93$, $p < 0.001$, CFI = 0.92, TLI = 0.90, RMR = 0.04, RMSEA = 0.09			
Chemical: $\chi^2 = 126.58$, $df = 38$, $p < 0.001$, CFI = 0.98, TLI = 0.97, RMR = 0.03, RMSEA = 0.07			

Note: CFI: comparative fit index; TLI: Tucker-Lewis index; RMR: root mean square residual; RMSEA: root mean square error of approximation.

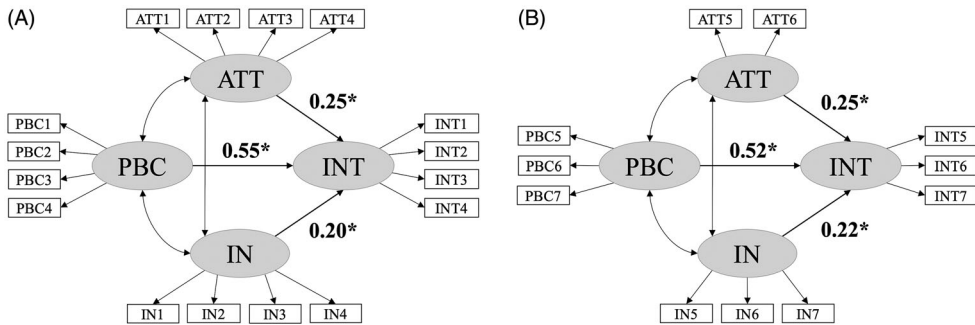


Figure 1. Full structural models (A = habitat; B = chemical) with the standardized estimates for the hypothesized relationships. Note: * $p < 0.001$; ATT: Attitudes, PBC: Perceived Behavioral Control, IN: Injunctive Norms, INT: Intentions. See Tables 1 and 2 for indicator descriptions.

of animal-mediated pollination; therefore, engagement approaches encouraging landholders to plant pollinator foraging habitat to promote cross-pollination of their crops will likely be ineffective. However, if a landholder had a favorable attitude toward pollinator habitat on their land for improving the health of the property, yet they are lacking knowledge or skills to implement BMPs, an effective approach could be developed, such as creating a skills-based landholder workshop. Targeted messaging and approaches are useful for identifying behavioral interventions and have the potential to improve conservation practice adoption (Floress et al. 2018; Marselle et al. 2020; Morrison et al. 2012; Pannell et al. 2006).

Our study also showed the importance of distinguishing between habitat and chemical practices when researching attitudes and normative pressures. Landholders may have an overall favorable or unfavorable attitude toward pollinator conservation but understanding their differentiation of the practices is crucial. The stigma surrounding the use of chemicals has been growing with the impetus to improve environmental quality and use non-

GMO crops. To this end, we found landholders differentiated between habitat and chemical indicators regarding the attitudinal and normative constructs. This finding indicates normative pressures may have more influence on intentions to adopt chemical rather than habitat BMPs. Thus, the promotional effectiveness of pollinator BMPs may vary on the type of practice and the landholder's antecedent belief associations.

There were four potential limitations of this study for which we have provided future research recommendations. First, we attempted to cover nine pollinator BMPs across several constructs, which resulted in similarities of indicator wording. To meet Fishbein and Ajzen's (2015) suggestion of indicators including the action, target, context, and time, the wording was kept as simple and similar as possible for the respondent. However, this wording may have influenced correlations and model fit. Future research could involve the development of more heterogeneous indicators. Additionally, the final chemical model had only two indicators for *Attitudes*, three for *Perceived Behavioral Control*, three for *Injunctive Norms*, and three for *Intentions*, leaving little room for model improvement. Creation of additional indicators would be beneficial to further explore this dimension. To avoid potential redundancy or ambiguity regarding *Attitudes* indicators in the future, semantic differential scales could be used with instrumental (e.g., "good-bad", "harmful-beneficial") or experiential (e.g., "enjoyable-unenjoyable", "ugly-beautiful") adjective pairs (see Fishbein and Ajzen 2015). See Westlake (2019) or [Supplemental Files](#) for the measurement instrument and indicator wording suggestions.

Second, the pollinator BMPs in this study were not all-inclusive and were active rather than passive practices. For instance, reducing mowing frequency can be beneficial to pollinator species, yet it is a more passive practice (i.e., the landholder has to do less or nothing at all; see Mader et al. 2011). We did not specifically ask landholders if fields were left fallow or if they let wildflowers grow, although many reported they did. These types of passive practices are beneficial, and their inclusion would further habitat quantity estimations if we knew where they were being used on the landscape and at what scale. Future research could address this by incorporating both active and passive pollinator BMPs in the measurement instrument to develop a better understanding of landholder support, or allowance, of pollinator foraging and nesting habitat on their properties.

Third, to address the relationships of the attributes, it was necessary to focus the landholder's responses to one property; however, this grounding of attitudes, normative pressures, constraints, and intentions meant we could not generalize their attributes beyond this specific property. Though the landholder may have owned multiple properties, we needed them to answer regarding the property about which we had information from the tax records. Recognizing this grounding and potential limitation, we asked if the landholders were using pollinator BMPs on any of their other landholdings and requested information about those properties. Future research should address how to obtain further information about the landholders' attributes affecting adoption as a whole without increasing the burden on the respondent.

Finally, although our results did provide evidence of reliability and validity, it was only one study. Further research is needed to continually test these indicators and potentially develop new ones to strengthen the findings. It would be useful to test this measurement instrument in various geographic locations, especially with landholders who more directly depend on the animal-mediated pollination ecosystem service, as it

would allow opportunities to examine metric invariance across different groups of individuals. Another important consideration for future research would be to use records that have additional demographic information, such as gender, race, and age, to conduct further non-response bias checks. In a recent article regarding declining survey response rates, researchers suggested it is crucial to both acknowledge the potential problems associated with low response rates and understand the respondent engagement (Stedman et al. 2019). In other words, are respondents those individuals in a population most interested in and involved with the topic? If so, the researcher should take care in interpreting the findings. Having additional demographic information and conducting follow-up phone interviews with non-respondents to determine the cause of their lack of response will help support the generalizability of results in future studies.

Conclusion

Our study was the first of its kind to look specifically at the influence of TPB attributes on pollinator BMP adoption intentions among private landholders. We aimed to address the gap of human dimensions research of pollinator conservation by understanding landholders' attitudes toward, normative pressures of, and constraints from adopting habitat and chemical BMPs on their property. Our study provided the development and testing of a measurement instrument that can be used and improved upon in future research, further incorporating human dimension aspects into pollinator conservation knowledge and efforts. The finding of *Perceived Behavioral Control* having the strongest relative influence on *Intentions* suggests the efficacy of pollinator conservation efforts would be improved with more targeted support for landholders to reduce feelings of constraint. To further increase habitat establishment and chemical reduction on private lands, we must better understand landholders. This will become more imperative as landholder generations and motivations continue to change. With the majority of US lands in private ownership, the future of pollinator conservation efforts is inextricably linked to understanding the vast wellspring of potential participants. More conservation social science research is needed to increase understanding and engagement, consequently bridging the colloquial research and Extension gap for the betterment of pollinator species and their habitat.

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